Physics 220 – Practice Final Exam 1

#1 - #15 **[2 points each]** Multiple Choice. Circle the letter of the correct answer.

1) An object of mass M is launched from ground height with an initial velocity of 200 m/s at an angle 30° above horizontal. To what maximum height will the object rise? Use g = 10 m/s/s. Ignore air resistance.

A) 20m
B) 50m
C) 250m
D) 500m
E) 2,000m

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2) In problem 1 above, what net force acts on the object when it reaches its highest point?

A) Zero
B) 173 N
C) g
D) 10 N
E) Mg

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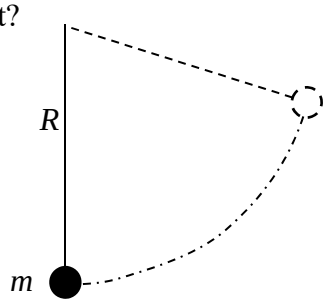
3) A boy exactly fills a 5-liter bucket with 4 equal shovelfuls of dirt. How many **cubic centimeters** of dirt are in each shovelful? Note: 1.0 m^3 = 1,000 liters

A) 80 cm^3 / shovelful
B) 125 cm^3 / shovelful
C) 800 cm^3 / shovelful
D) 1250 cm^3 / shovelful
E) 8,000 cm^3 / shovelful

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4) A ball of mass m, attached to a lightweight string of length R, is dropped from a certain height. The other end of the string is attached to a pivot. If the ***tension*** at the bottom-most point of the swinging motion is equal to $2mg$, what is the speed, v, of the ball at that instant?

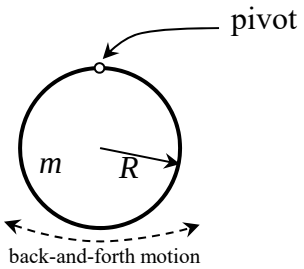
A) $\sqrt{Rg}$
B) 20m
C) Rg
D) Zero
E) $\sqrt{3Rg}$



---

5) Determine the **period** of a pendulum that consists of a uniform sphere of mass m and radius R. The sphere pivots about a small steel hinge mounted at a position on the sphere's edge. Assume it pivots with a small angle. For a sphere, $I_{c.o.m.} = {}^2/_5\, m\, R^2$

A) $2\pi\sqrt{2R/5g}$
B) $2\pi\sqrt{R/g}$
C) $2\pi\sqrt{2R/g}$
D) $2\pi\sqrt{7R/5g}$
E) $2\pi\sqrt{3R/5g}$



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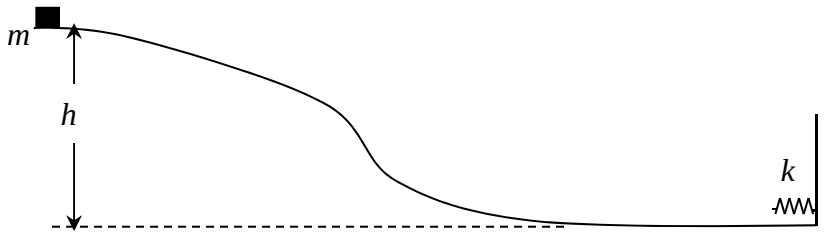
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A) $\pi/3$
B) 1/3
C) 3
D) $3/\pi$
E) More information needs to be given.

---

7) A mass *m* slides from rest down a frictionless hill of height h. At the bottom the mass compresses a lightweight spring by 3 meters before stopping for an instant. What is the spring constant of the spring?

A) $\frac{2}{3}mgh$
B) $\frac{1}{3}mgh$
C) $\frac{2}{9}mgh$
D) $\frac{1}{3}mgh$
E) $\sqrt{\frac{2}{3}}mgh$



---

8) If the mass of Pluto is $1x10^{22}$ kg and the radius of pluto is 1200 km, what is the approximate escape speed from the surface of Pluto? ($G = 6.7x10^{-11}$ $m^3/(kg \cdot s^2)$).

A) 0.1 km/s
B) 0.3 km/s
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9) Escape velocity from earth is 11 km/s. What is the minimum energy it would it take to launch a 1.0 gram object with that escape velocity? Ignore air resistance.

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10) A 5.0 kg block traveling with a velocity 4m/s to the north crashes into a stationary 3.0kg block in an ***elastic*** collision. How much mechanical energy is lost to the system during the collision? Consider both blocks to be part of the system.

A) Zero
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D) 16 J
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11) A round disk of mass M=1kg and radius R=1meter is accelerated about its center of mass. From rest, it is spun up to its final angular speed of 20 rad/s with a constant angular acceleration. If it takes 2 seconds for the disk to get to its final speed, what is the angular acceleration during that time? The moment of inertia of a disk is ½ MR^2

A) 1 rad/s^2
B) 2.2 rad/s^2
C) 4.5 rad/s^2
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E) 40 rad/s^2

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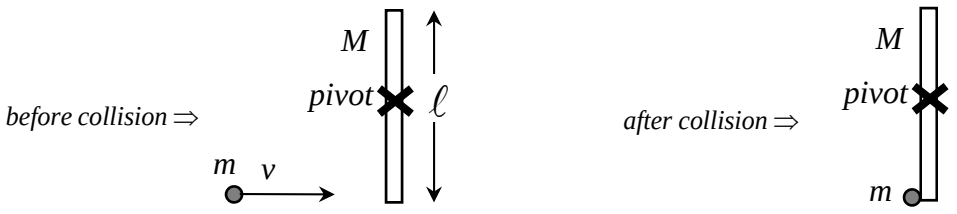
12) In problem 11 above, how much mechanical energy does the disk have when it is rotating at its final speed?

A) 0.25 J
B) 100 J
C) 200 J
D) 1,000 J
E) 4,000 J

---

13) A gum ball of mass *m* is fired with a velocity *v* at a ***stationary*** thin rod of length ℓ and mass *M* that is free to pivot about its center. The gum ball strikes the end of the rod and sticks-in-place as shown. What is the angular momentum of the rod+gum system immediately after the collision? $I_{rod\ c.o.m.} = \frac{1}{12} M\ell^2$

A) $\frac{1}{2} mv\ell$
B) $(\frac{1}{2} m + \frac{1}{12} M)v\ell^2$
C) $(\frac{1}{2} m + \frac{1}{12} M)v\ell$
D) $m + \frac{1}{12} M\ell^2$
E) $vmM\ell/(m+M)$



---

14) The position of an oscillating mass-on-a-spring is described by the following equation: $x(t) = 0.2 \cos(2\pi\, t + \pi/2)$. What is the ***acceleration*** of the mass at time $t = 0.5$s ? (assume SI units).

A) 0
B) $4\pi^2$
C) $0.8\pi^2$
D) 0.4π
E) 2π

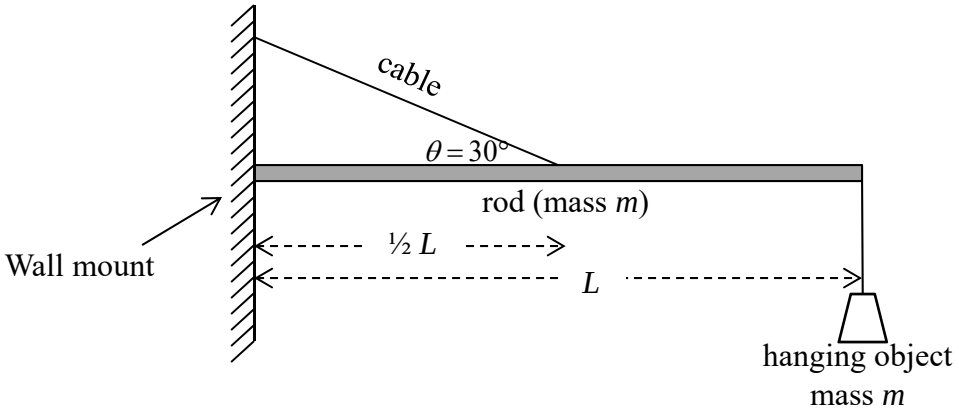
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15) If you were to take this class for the first time and only use a digital copy of the text (a 'pdf' copy), what would your opinion be? (all honest answers receive credit)

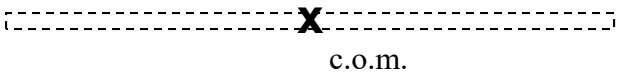
A) Having **only** a pdf copy of the text is fine.
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D) It doesn't matter, I barely look at the text anyway.
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16) (10 pts) A uniform rod of length *L* and mass *m* is supported by a wall-mount and a cable as shown. An object also of mass *m* hangs from the rod at the end. Everything remains at rest.



A) (4 pts) Draw the extended free body diagram for the rod. Indicate the location and approximate direction of all forces acting on the rod. Label them with conventional labels.



B) (3 pts) Using the point where the rod contacts the wall as the pivot point, write the sum of all of the torques directed into-the-page, and separately all of the torques directed out-of-the-page. **Your answers should be written in terms of *L*, *m*, *g*, and the cable tension *T*.**

$\tau_{\otimes}$ = _____ (*into the page*)

$\tau_{\odot}$ = _____ (*out of the page*)

C) (1 pts) Solve for the cable tension T in terms of *L*, *m*, and *g*.

T_{cable} = _____

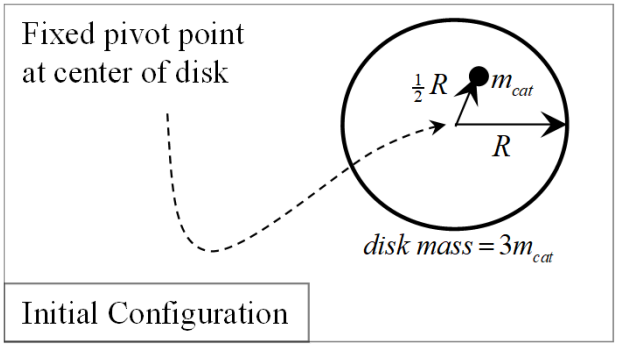
D) (2 pts) In terms of L, m, and g what is the vertical component of the force that the wall exerts on the rod? Include the **direction!**

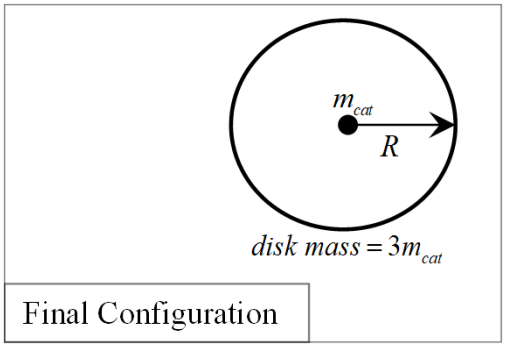
$F_{\substack{vert \\ wall}}$ = _____; *direction* _____

Physics 220 – Practice Final Exam 1

17) (10 pts) A cat stands on a horizontal rotating solid disk of radius *R* halfway to the edge. Assume the cat acts like a point mass at a distance ½ R from the pivot point. The cat has a mass *m*, and the disk has 3-times the mass of the cat (so the system has a total mass of 4*m*). The cat+disk system rotates initially with an angular speed of 2.0 rad/sec. The cat then walks to the center of the disk.
Note: $I_{initial}$ and I_{final} are already given in part (A).

$General: solid\ disk\ of\ mass\ M, radius\ R \Rightarrow I_{disk} = \frac{1}{2}MR^2$





Initially **the system rotates with |ω|=2.0 rad/sec**.

The cat walks to the center of the disk (exactly on the pivot point). Assume the cat still acts like a point mass.

A) (5pts) Fill in the boxes below, ***your answers must be in terms of m and R***

Before cat moves

$I_{cat+disk} = \frac{7}{4} m_{cat} R^2$

$|\omega_i| = 2.0 \frac{rad}{s}$

$|L_i| =$ _____

$K.E._i =$ _____

After cat moves to center

$I_{\substack{system\\about\\pivot}} = \frac{3}{2} m_{cat} R^2$

$|\omega_f| =$ _____

$|L_f| =$ _____

$K.E._f =$ _____

B) (2pts) **Circle** the correct relationship:

$KE_i > KE_f$ $\qquad$ $KE_i = KE_f$ $\qquad$ $KE_i < KE_f$

C) (3 pts): Show that the ***initial*** moment of inertia of the system is $I_{cat+disk} = (7/4)\, m_{cat} R^2$;
Start with $I_{disk} = \frac{1}{2} MR^2$ Show all work!

Physics 220 – Practice Final Exam 1

18) A) (6 points) Draw the first 3 resonant modes for standing sound waves in a hollow pipe with one closed end. Since sound waves are compression waves which are difficult to draw, you should represent the sound waves as transverse waves to illustrate where the ***displacement*** nodes and anti-nodes occur.



Fundamental mode
(1st Harmonic)



Next possible mode
(3rd Harmonic)



Next possible mode
(5th Harmonic)

B) (2 pts) The pipe above is has a fundamental frequency f_1 = 100 Hz. If the speed of sound is 340 *m*/*s*, what is the length of the pipe?

C) (2 pts) What is the spacing between nodes for the 5th harmonic?

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A) 20*m*
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2) In problem 1 above, what net force acts on the object when it reaches its highest point?

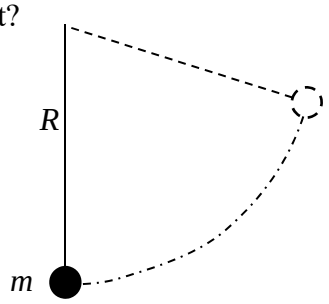
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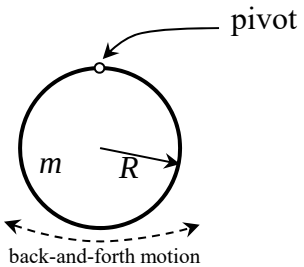
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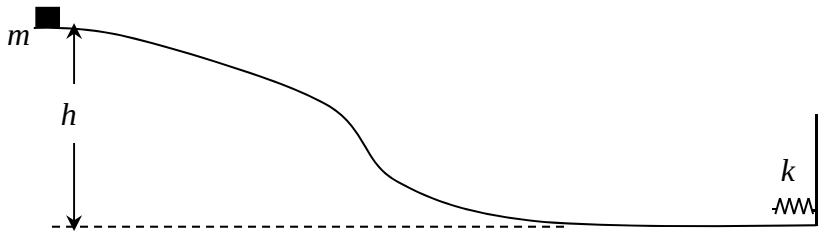
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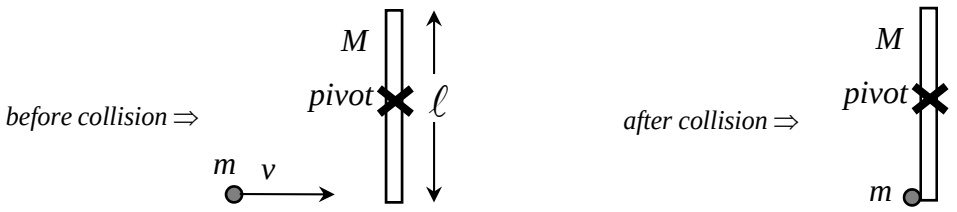
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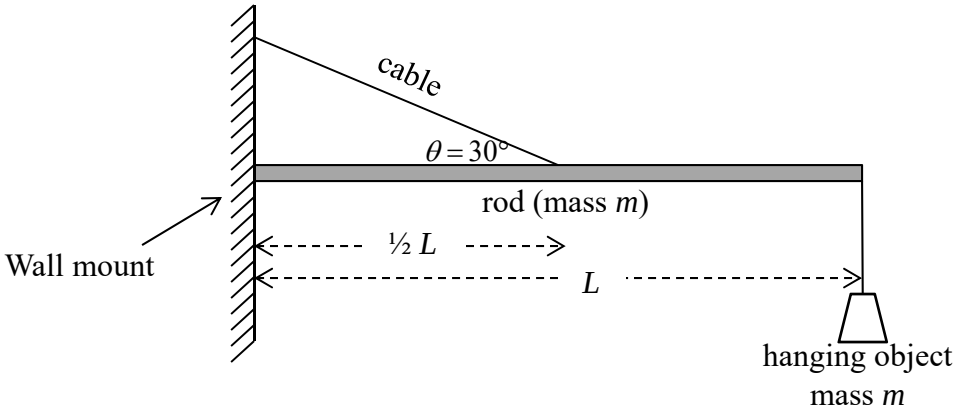
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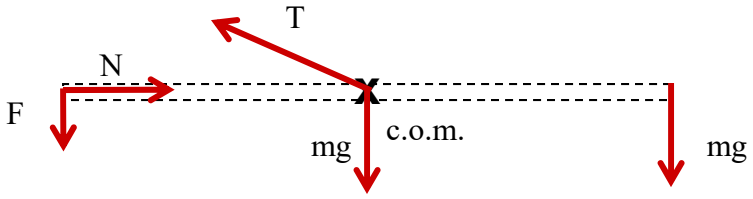
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$\tau_{\otimes} =$ $\frac{L}{2}mg + Lmg = \frac{3}{2}Lmg$ *(into the page)*

$\tau_{\odot} =$ $\frac{L}{2}T\sin 30 = \frac{1}{4}LT$ *(out of the page)*

C) (1 pts) Solve for the cable tension T in terms of *L*, *m*, and *g*.

$$\tau_{in} = \tau_{out}$$
$$\tfrac{1}{4}LT = \tfrac{3}{2}Lmg$$
$$T = \tfrac{12}{2}mg$$
$$T = 6mg$$

$T_{cable} =$ $6mg$

D) (2 pts) In terms of L, m, and g what is the vertical component of the force that the wall exerts on the rod? Include the **direction!**

$$F_{net\,y} = 3mg - mg - mg - F_{\substack{vert \\ wall}} = 0$$

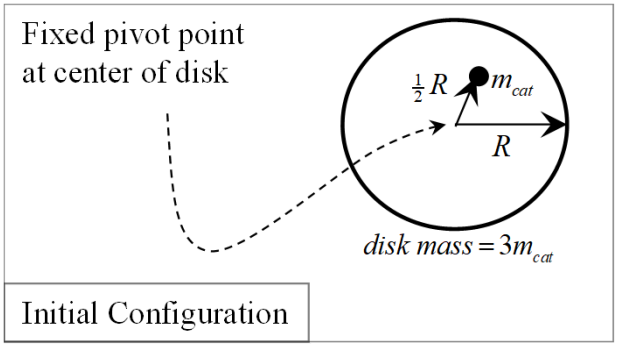
$$F_{\substack{vert \\ wall}} = mg$$

$F_{\substack{vert \\ wall}} =$ mg; direction $down!$

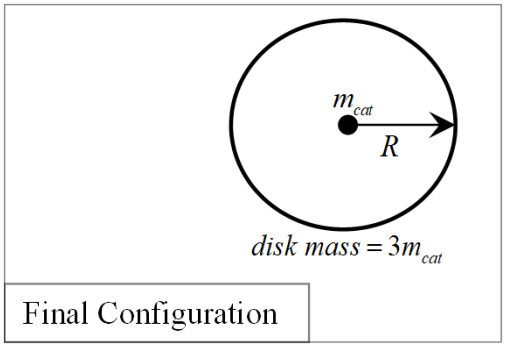
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Initially **the system rotates with |ω|=2.0 rad/sec**.



The cat walks to the center of the disk (exactly on the pivot point). Assume the cat still acts like a point mass.

A) (5pts) Fill in the boxes below, ***your answers must be in terms of m and R***

Before cat moves

$I_{cat+disk} = \frac{7}{4}m_{cat}R^2$

$|\omega_i| = 2.0\ \frac{rad}{s}$

$|L_i| =$ $\frac{7}{2}mR^2$

$K.E._i =$ $\frac{7}{2}mR^2$

$L_i = I_i\omega_i$

$L_i = \frac{7}{4}mR^2(2)$

$= \frac{7}{2}mR^2$

$KE_i = \frac{1}{2}I\omega_i^2$

$KE_i = \frac{1}{2}\frac{7}{4}mR^2(2)^2$

$= \frac{7}{2}mR^2$

$L_f = I_f\omega_f$

$L_f = \frac{3}{2}mR^2\omega_f$

$\frac{7}{2}mR^2 = \frac{3}{2}mR^2\omega_f$

$\frac{7}{3} = \omega_f$

$KE_f = \frac{1}{2}I\omega_f^2$

$KE_f = \frac{1}{2}\left(\frac{3}{2}mR^2\right)\left(\frac{7}{3}\right)^2$

$= \frac{49}{12}mR^2$

After cat moves to center

$I_{\substack{system\\about\\pivot}} = \frac{3}{2}m_{cat}R^2$

$|\omega_f| =$ $\frac{7}{3}$

$|L_f| =$ $L_i = \frac{7}{2}mR^2$

$K.E._f =$ $\frac{49}{12}mR^2$

B) (2pts) **Circle** the correct relationship:

$KE_i > KE_f$ $\quad$ $KE_i = KE_f$ $\quad$ (**$KE_i < KE_f$**)

C) (3 pts): Show that the ***initial*** moment of inertia of the system is $I_{cat+disk} = (7/4)\, m_{cat} R^2$;
Start with $I_{disk} = \frac{1}{2}MR^2$ Show all work!

$I_{tot} = I_{cat} + I_{disk}$

$= m(\frac{R}{2})^2 + \frac{1}{2}(3m)R^2$

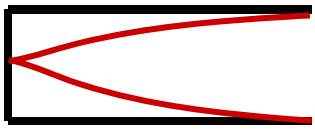
$= (\frac{1}{4} + \frac{3}{2})mR^2$

$= (\frac{1}{4} + \frac{6}{4})mR^2$

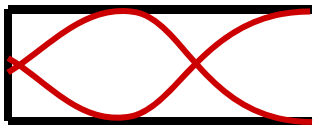
$= \frac{7}{4}mR^2$

Physics 220 – Practice Final Exam 1

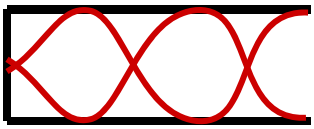
18) A) (6 points) Draw the first 3 resonant modes for standing sound waves in a hollow pipe with one closed end. Since sound waves are compression waves which are difficult to draw, you should represent the sound waves as transverse waves to illustrate where the ***displacement*** nodes and anti-nodes occur.



Fundamental mode
(1st Harmonic)



Next possible mode
(3rd Harmonic)



Next possible mode
(5th Harmonic)

B) (2 pts) The pipe above is has a fundamental frequency f_1 = 100 Hz. If the speed of sound is 340 *m*/*s*, what is the length of the pipe?

$$\tfrac{1}{4}\lambda_1 = L \quad \Rightarrow \quad \lambda_1 = 4L$$
$$\tfrac{3}{4}\lambda_3 = L \quad \Rightarrow \quad \lambda_3 = \tfrac{4}{3}L$$
$$\tfrac{5}{4}\lambda_5 = L \quad \Rightarrow \quad \lambda_5 = \tfrac{4}{5}L$$

$$v = \lambda_1 f_1$$
$$v = 4Lf_1$$
$$\frac{v}{4f_1} = L = \frac{340}{4(100)}$$

$$L = 0.85m = 85cm$$

C) (2 pts) What is the spacing between nodes for the 5th harmonic?

$$\lambda_5 = \tfrac{4}{5}L = \tfrac{4}{5}(0.85)$$
$$\lambda_5 = 0.68$$

spacing d between nodes is one – half wavelength

$$d = 0.34cm$$